



INF212 ALGORITHMS AND PROGRAMMING

PROJECT-3 Drone Delivery System Automation

The deadline is Wednesday, May 04, 2026, at 23:59.

Projects that are not delivered on time are not accepted.

Upload the project to the Project 3 assignment section of the INF212 class.

The questions can be asked to course lecturer Dr. Murat CEYLAN and teaching assistant Dr. Murat Enes HATİPOĞLU.

Upload your files as followings:

- After uploading your source files, you must use **TURN IN** button (**GÖNDER** butonu) in the system, otherwise it will be a **draft** of your version and will not be accepted.
- MS Teams "**INF212**" class is the directory for your files.
- Do not create a folder or RAR or whatever, just **UPLOAD your source codes**.
- Prepare a project report using "INF212_Project_Report_Template.docx".

PROJECT 3

In this project, it is expected to develop an autonomous drone delivery system in C language with the help of product, customer and storehouse information. Product purchase information should be analyzed and displayed according to customer location, product type, quantity and nearest storehouse locations. Data will be kept using the single linked list feature.

Data structures in this project:

- Product: ID, name, type, quantity, price
- Customer: ID, name, location (x, y)
- Storehouse: ID, location (x, y)

Each structure must consist of a linked list feature separately.

The drone delivery system automation menu can be designed as follows:

1. Add Product
2. Add Customer
3. Add Storehouse
4. View Products
5. View Customers
6. View Storehouses
7. Analyze Purchase
8. Exit

1. **Add Product:** The product list is created by a user with this selection. Product structure members (ID, name, type, quantity and storehouse ID) are entered by the user. Added products are listed with the linked list feature.
2. **Add Customer:** The customer list is created by a user with this selection. Customer structure members (ID, name, location (x, y)) are entered by the user. Added customers are listed with the linked list feature.
3. **Add Storehouse:** The storehouse list is created by a user with this selection. Storehouse structure members (ID, location (x, y)) are entered by the user. Added storehouses are listed with the linked list feature.
4. **View Products:** All the product information is printed with this selection.
5. **View Customer:** All the customer information is printed with this selection.
6. **View Storehouse:** All the storehouse information is printed with this selection.
7. **Analyze Purchase:** In this selection, a user selects a customer from the created customer list, then selects a product from the created product list, finally enters the amount of the selected product. Then, the algorithm finds the nearest storehouse, analyzes the product availability and calculates the total price and the remaining product amount. A shipping fee is calculated according to the distance and is added to the total price. At the end, customer information (name, location), product information (name, type, price), product

amount, nearest storehouse information (ID, location), remaining product amount, total product price, shipping fee (5 TL per kilometer), and total price are listed as an output.

8. **Exit:** Program is closed with this selection.

The use of structures, dynamic memory allocation, and pointers is mandatory.

Preparing the project report using 'INF212_Project_Report_Template.docx' is mandatory.

Customer structure is given below as a guidance:

```
typedef struct Customer {
    int id;
    char name[50];
    float x, y;
    struct Customer *next;
} Customer;

Customer *customerHead = NULL;
```

For testing the program, the data in the tables below can be used.

Customer ID	Customer Name	Address
101	Ali Karaman	x = 1, y = 5
102	Zehra Cilek	x = 6, y = 0
103	Mehmet Derin	x = 3, y = 4
104	Veli Ortanca	x = 1, y = 11
105	Menekse Serin	x = 5, y = 7

Product ID	Product Name	Product Type	Product_Price
201	Portakal Sıkma (1 kg)	Manav	24,95 TL
202	Muz Yerli (1 kg)	Manav	114,85 TL
203	Tükenmez kalem	Kirtasiye	50 TL
204	Karton A4 kâğıt	Kirtasiye	500 TL
205	Nohut (1 kg)	Temel Gıda	65,95
206	Baldo pirinç (1 kg)	Temek Gıda	60,45

Storehouse ID	Address
301	x = 0, y =4
302	x = 3, y =2
303	x = 8, y = 9
304	x = 10, y = 4